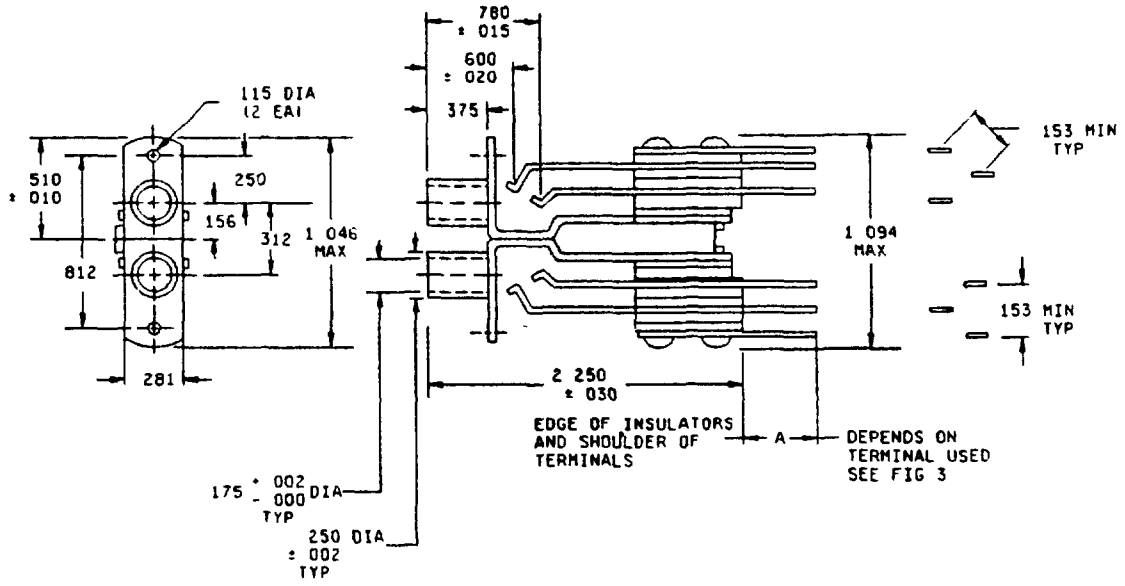
INCH-POUND
MIL-J-641/21A
12 November 1993
SUPERSEDING
MIL-J-641/21
26 February 1979

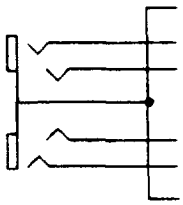
MILITARY SPECIFICATION SHEET

JACKS, (BANTAM) TELEPHONE, 3 CONDUCTOR, DOUBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-J-641.





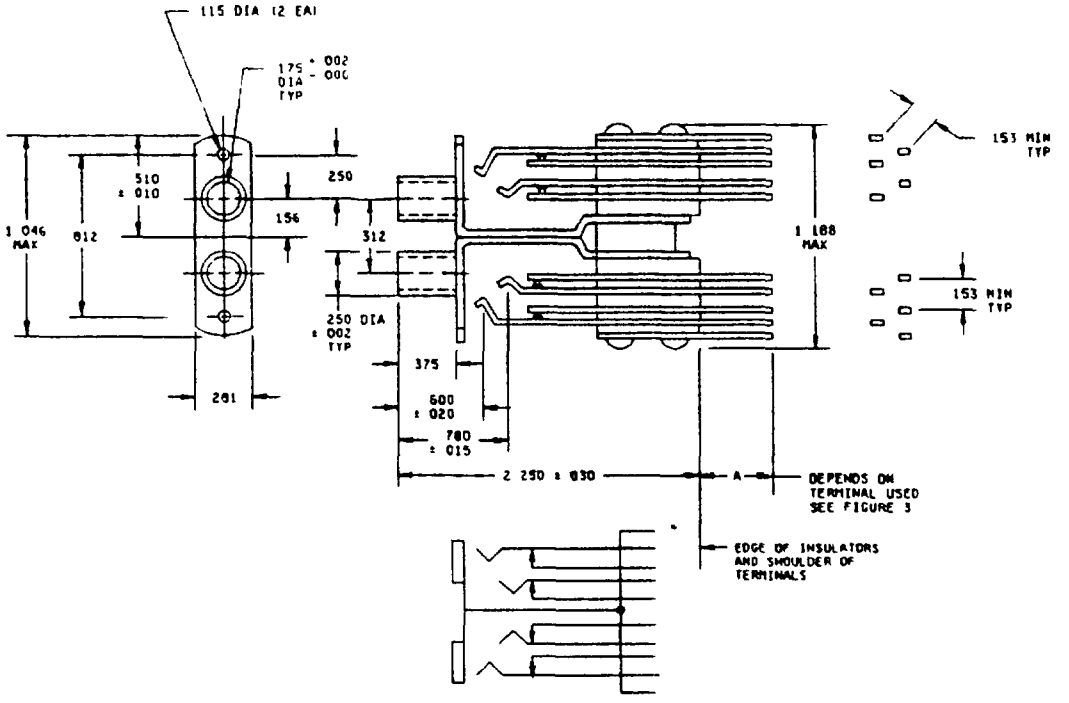
CIRCUIT DIAGRAM

Dash nos -1 and -2 (no breaks)

FIGURE 1 Dimensions and configurations

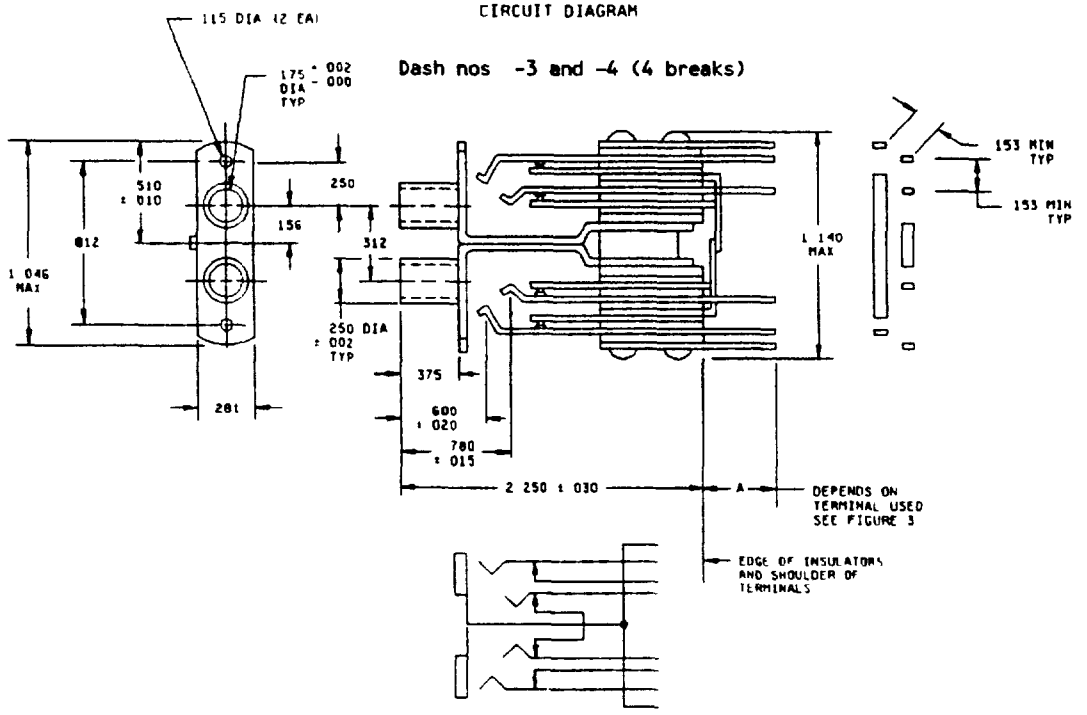
DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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CIRCUIT DIAGRAM

Dash nos -3 and -4 (4 breaks)

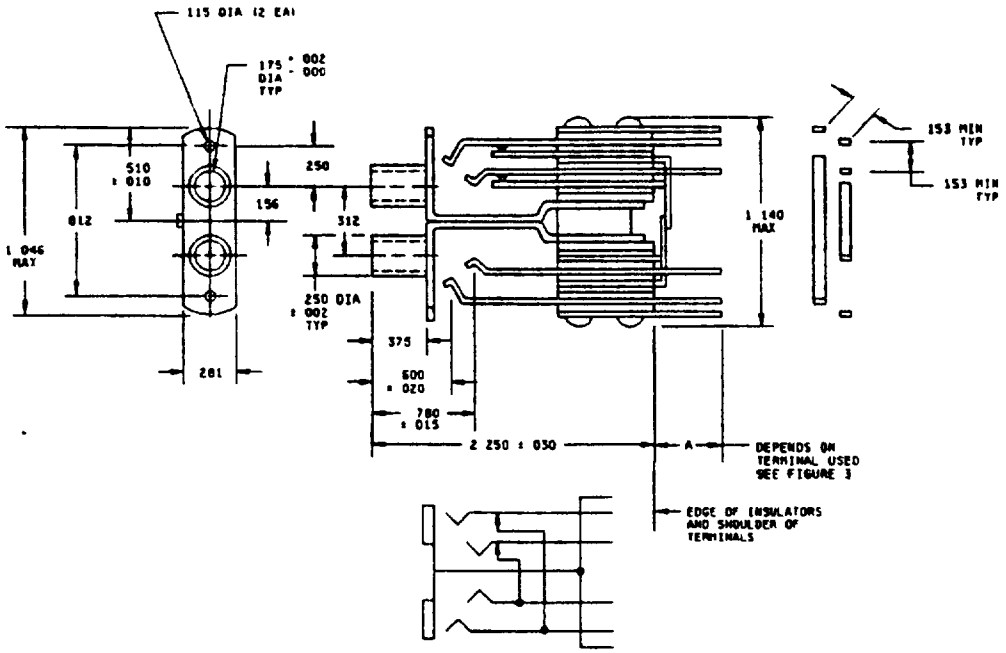


CIRCUIT DIAGRAM

Dash nos -5 and-6 (4 breaks connected)

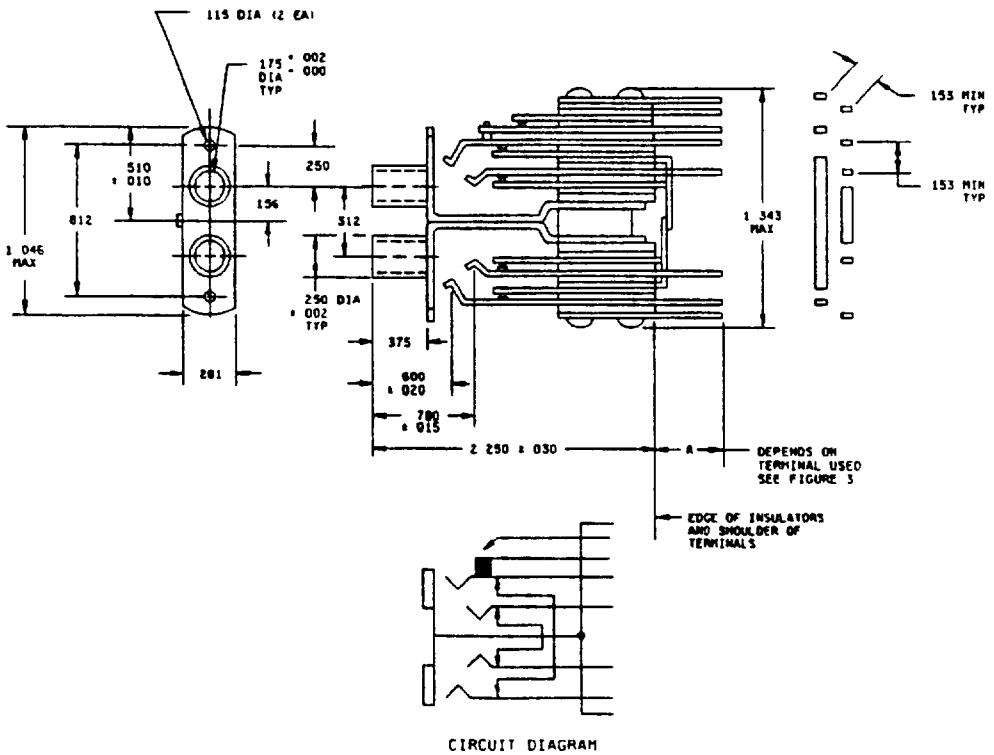
FIGURE 1 Dimensions and configurations - Continued

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CIRCUIT DIAGRAM

Dash nos. -7 and -8 (2 breaks connected)

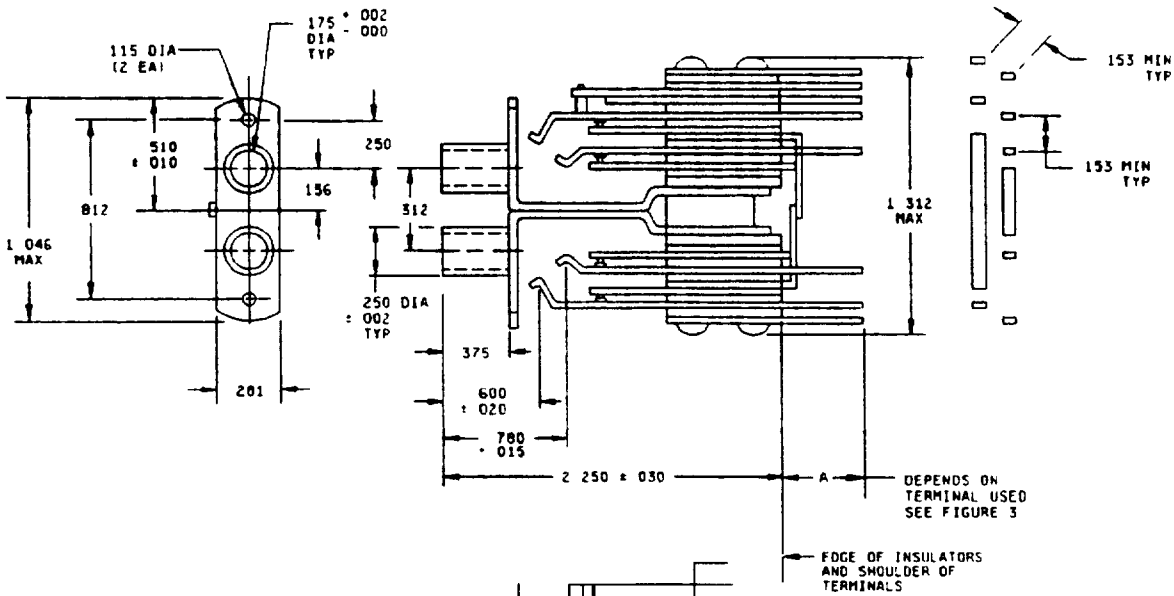


CIRCUIT DIAGRAM

Dash nos -9 and -10 (4 breaks connected, 1 make)

FIGURE 1 Dimensions and configurations - Continued

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CIRCUIT DIAGRAM

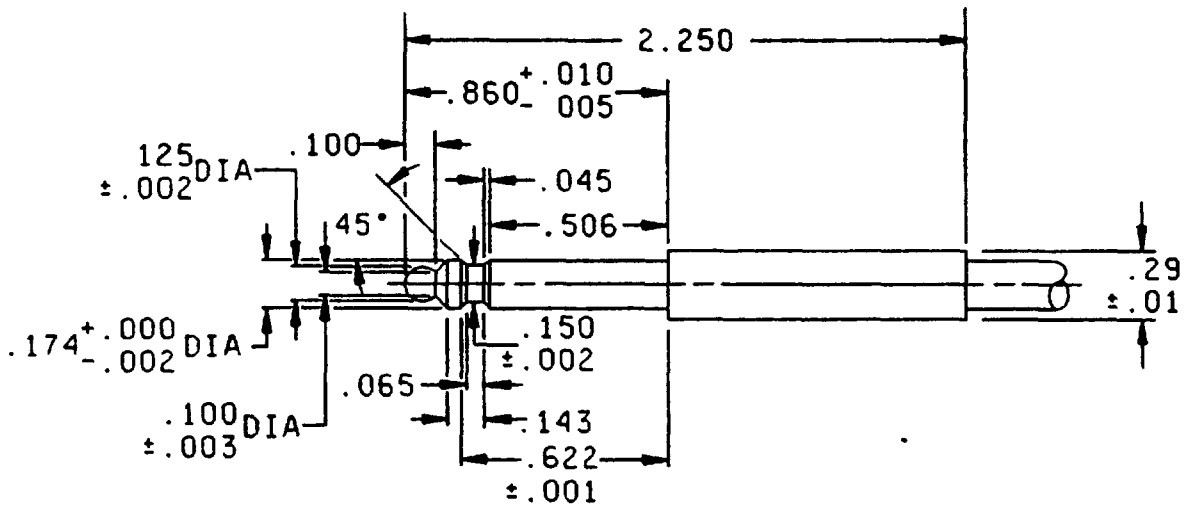
Dash nos -11 and -12 (5 breaks, 4 connected)

Inches	mm	Inches	mm	Inches	mm
002	0.05	.175	4 44	812	20 62
.010	0.25	250	6 35	1 046	26 57
.015	0.38	.281	7 14	1 094	27 79
020	0 51	312	7 92	1 140	28 96
030	0 76	375	9.52	1 188	30 18
115	2.92	.510	12.95	1.312	33.32
153	3 89	.600	15 24	1 343	34 11
156	3 89	780	19 81	2 250	57 15

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerance is ± 005 (0 13 mm)
4. All dimensions are after plating
5. Wire shunts are permitted
6. All undimensioned pictorial features are for reference purposes only
7. For wirewrap type jacks, terminal and contact interconnecting jumpers shall not interfere with commonly used wirewrap tools

FIGURE 1 Dimensions and configurations - Continued

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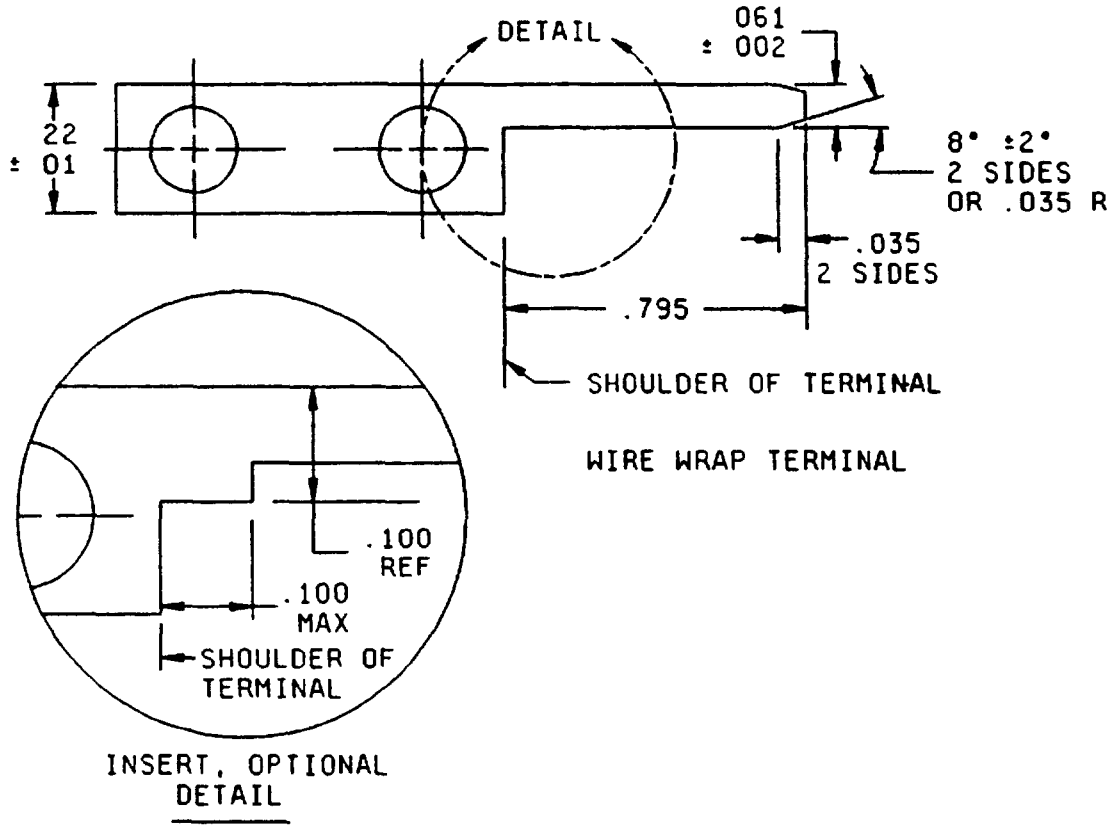
Inches	mm	Inches	mm	Inches	mm
.001	0 03	065	1 65	.29	7.4
002	0 05	.100	2 54	.506	12 85
003	0 08	125	3.18	.622	15 80
005	0 13	143	3 63	860	21 84
010	0 25	150	3 81	2.250	57 15
045	1 14	174	4 42		

NOTES

1 Dimensions are in inches
2 Metric equivalents are given for information only
3 Unless otherwise specified, tolerances are ±.005 (0.13 mm) and 1° on angles.
4 The material shall be brass
5 Finger shall be hardened to Rockwell C 60-62, polished, and then hard-chromium plated to 0003 inch (0 008 mm) maximum thickness.
6 All dimensions are after plating.
7 Concentricity shall be 003 (0 08 mm) total indicator reading

FIGURE 2 Test plug TP-777

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Inches	mm	Inches	mm
002	0 05	100	2.54
01	0 3	22	5 6
035	0 89	795	20 19
061	1 55		

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerance is ± 005 (0 13 mm)
4. Solder lug terminals (not shown) shall meet the requirements of MIL-J-641, 3.5 5, and the "A" dimension of figure 1 shall be 325 inch (8 26 mm) maximum
5. All dimensions are after plating

FIGURE 3 Terminals

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REQUIREMENTS

Dimensions and configurations See figure 1

Terminals See figure 3 (wirewrap) or MIL-J-641 (solder lug)

Materials
  Springs Nickel silver.
  Contact Palladium or gold alloy

Insertion and withdrawal forces
  Insertion force 7 pounds maximum
  Withdrawal force 1 0 pound minimum, 7 pounds maximum
  Test plug TP-777 (see figure 2)

Spring life Test plug TP-777 shall be used

Frame strength. Frame strength test is applicable Load - 7 pounds maximum

Terminal of tip contact spring shall be on the right side when terminals are viewed from the rear, with the jack in a position such that the contact spring pileup is on top of the jack frame

Cutout in the jack frame shall be positioned so that a minimum of .016-inch clearance exists between the tip of the TP-777 plug and the edge of the frame cutout at all times during insertion of the plug

All terminals shall be stacked as shown on schematic (see figure 1)

Part or Identifying Number (PIN) M641/21 - (dash number from table I)

Mating plugs See table II

TABLE I Dash numbers and characteristics

Dash number	Connection	Mounting	Switching action	Terminal
-1	Tip, ring, and sleeve	Double lug	None	Wirewrap
-2	"	"	None	Solder lug
-3	"	"	4 breaks	Wirewrap
-4	"	"	4 breaks	Solder lug
-5	"	"	4 breaks connected	Wirewrap
-6	"	"	4 breaks connected	Solder lug
-7	"	"	2 breaks connected	Wirewrap
-8	"	"	2 breaks connected	Solder lug
-9	"	"	4 breaks connected, 1 make	Wirewrap
-10	"	"	4 breaks connected, 1 make	Solder lug
-11	"	"	5 breaks, 4 connected	Wirewrap
-12	"	"	5 breaks, 4 connected	Solder lug

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TABLE II Mating plugs

Dash number	Mating plug
-1	M642/13-1, -2, -3, -4
-2	"
-3	"
-4	"
-5	"
-6	"
-7	"
-8	"
-9	"
-10	"
-11	"
-12	"

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities
Army - AR, AV, ME, MI
Navy - AS, MC, SH
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3910-03)

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